

Working Around the “Smart” Classroom: Teachers and On-site Staff as Infrastructure Repair Workers in a Chinese K–12 EdTech Deployment

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Vendor-managed “smart classroom” and “AI-augmented” systems are often evaluated through an adoption frame: whether teachers log in, use features, and generate dashboard-visible activity. Drawing on a retrospective qualitative corpus generated during employment with one Chinese educational-technology vendor—11 interviews or brief elicitation records, 5 classroom observations, and 28+ days of field notes across eight focal schools and related site visits—I show that adoption metrics obscure a different phenomenon: teachers and on-site delivery staff perform ongoing **infrastructure repair labor** [29, 50] to make the deployed suite work in classroom practice. I identify four forms of this labor: **temporal stitching**, **defensive redundancy**, **content re-verification**, and **mediation labor**. I then show that repair labor becomes unevenly visible across **vendor-attention tiers**: breakdowns at flagship sites are more likely to become legible to staff and product repair channels, while similar breakdowns at lower-attention sites are more often absorbed locally. The paper’s primary contribution is an empirical account of repair labor in a Chinese K–12 EdTech deployment. Building on that account, it develops the CSCW concept of *visibility asymmetry*, extending infrastructure theory by showing that breakdowns become visible not simply when they occur, but when organizational repair channels can carry them upward.

CCS Concepts: • **Human-centered computing** → **Empirical studies in HCI**; **Collaborative and social computing theory, concepts and paradigms**.

Additional Key Words and Phrases: teacher-centered AI, infrastructure, repair, adoption, K–12 education, China, critical EdTech, design ethics, HCI

1 Introduction

It is 2:00 p.m. at School-S1, a flagship deployment site in a provincial-capital region. TeacherT06, a high-school math teacher, opens the school-deployed homework platform and begins what she calls “picking questions.” She has roughly three and a half hours before evening self-study begins. The platform’s print-and-deliver service promises a four-hour turnaround; she has negotiated with the on-site delivery staff for an emergency window. As on most days, she overrides the platform’s question recommendations—“the questions it pushes are not necessarily the ones I need”¹—and selects items manually from the legacy question bank that predates the deployment. The platform’s “layered” difficulty tags do not match how her subject group builds assessments; she disables the tagging and re-sorts by question type. Once or twice a week, she finds that a published assignment has vanished from her view and asks staff to print extra copies so she can recover the answer key. None of this appears in the platform’s usage dashboards. All of it appears in her working afternoons.

¹TeacherT06, Module-H teacher interview, 00:37.

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In another classroom, a sixth-grade writing lesson opens with the teacher scanning a QR code to log into the classroom-interaction module; the code is too large for the tablet's viewfinder, and she steps into the student desks to scan successfully.² The lesson opens to the wrong page; she flips manually. Algorithmic student selection repeatedly picks students who cannot answer; she abandons it for show-of-hands.³ An embedded video later renders at unreadable resolution. After class she reports that carrying the tablet through the lesson is exhausting and that the student-roster feature is buried too deep to find.

Industry discourse describes what TeacherT06 and her primary-school colleague do as successful **adoption**: high-frequency engagement with platform workflows at active sites. EdTech research in the Technology Acceptance tradition would code these practices similarly, asking which features are useful, which barriers persist, and how behavior compares to a target usage profile. Both framings are not wrong, but they are flat. They miss labor generated by gaps between the platform's model of classroom work and the rhythms, constraints, and pedagogical judgments that make a Chinese K–12 classroom run. They also miss that this labor is **unevenly routed**: even high-attention sites require substantial repair, while lower-attention sites more often absorb comparable breakdowns locally. They further miss that this labor is **shared**: the vendor's on-site staff—the embedded employees who run print-and-deliver cycles, plug in tablets every morning, and broker between teachers and the product team—are also performing repair work invisible in the dashboard.

This paper re-reads teachers' and on-site staff members' everyday engagement with deployed K–12 EdTech in China through an **infrastructure-and-repair** lens drawn from Star and Ruhleder [50], Star [49], and Jackson [29]. Rather than as adopters, I read these actors as **repair workers**: agents whose skilled, tacit, ongoing labor sustains the appearance that the deployed system “works.” I treat “AI-augmented,” “smart classroom,” and “intelligent grading” labels as part of this visibility regime: they overstate the deployed stack's pedagogical capabilities and help render human repair labor analytically invisible.

Guided by this framing, the paper asks two questions: **RQ1**: What repair labor do teachers and on-site staff perform around deployed K–12 EdTech? **RQ2**: How do organizational routing conditions shape which breakdowns become visible to product teams?

Empirically, I provide a multi-site, multi-module description of teacher and on-site staff repair labor around a deployed Chinese K–12 EdTech suite. **Conceptually**, I propose *visibility asymmetry across vendor-attention tiers* as a structural feature that shapes which breakdowns become product-visible and which are offloaded to teachers and staff. **For CSCW**, the paper analyzes the deployed suite as a cooperative infrastructure sustained by teachers, administrators, on-site staff, and product teams, rather than as a tool used by isolated teachers.

2 Related Work

2.1 Teacher Adoption as the Default Frame

The deployment of educational technology in K–12 classrooms—AI-marketed or otherwise—has been accompanied by a robust evaluative vocabulary organized around teacher **adoption**: whether, how frequently, and with what attitudes teachers take up a system. This vocabulary draws on the Technology Acceptance Model [13] and the Unified Theory of Acceptance and Use of Technology [57]. In industry discourse, it becomes dashboard practice: adoption rates, feature frequencies, monthly active users, and session lengths become the primary surfaces through which product teams “see” teacher behavior. Learning-analytics scholarship has shown that dashboards are most useful when embedded in situated teaching practice and designed with teacher stakeholders rather

²Observation 4.

³Observation 3.

than treated as neutral mirrors [45, 59]. That distinction is central here: the Vendor dashboard is not merely a pedagogical support tool but also an organizational visibility instrument.

This framing enables comparison across sites, identification of features failing to take up, and targeted teacher training. It also comes with three defaults the data complicate: it treats **technology as a stable object** that teachers take up or resist; it **locates friction in the teacher** rather than in the technology or its institutional surround; and it is **structured around feature-level visibility**, so practices not registered on the dashboard disappear analytically.

A related HCI thread articulates a *teacher-centered AI* alternative. Holstein and Alevén [26]; Holstein, McLaren, and Alevén [27]; Han et al. [23]; and Harvey, Koenecke, and Kizilcec [24] argue for designs that work **with** teachers’ pedagogical judgment, student agency, and classroom rhythms. This paper extends that discussion into a Chinese K–12 deployment context, names infrastructure repair as a mechanism the adoption frame misses, and adds on-site staff to the analysis.

2.2 Infrastructure Studies

Infrastructure studies in CSCW and STS provide the conceptual vocabulary for this move. Star and Ruhleder [50] define infrastructure **relationally**: infrastructure exists “in relation to organized practices,” and is often invisible in use, visible upon breakdown. Star [49] turns this into a method: study invisible work, installed bases, and the maintenance and coordination that hold systems together. Bowker and Star [4] extend this to **classification infrastructures**: categories built into a system pre-figure what can be seen, counted, and acted upon. This point returns when I read the Vendor’s dashboards as classification infrastructure. Pipek and Wulf’s [40] *infrastructuring* reframes the unit of analysis as the ongoing work that turns artifacts into infrastructure-in-use, and Plantin et al. [41] bridge infrastructure studies with platform studies to analyze commercially-managed digital systems—a frame I extend to vendor-deployed K–12 platforms in §4.5.

A related thread—articulation work—develops the same insight from a labor angle. Strauss [52, 53] defines articulation work as the effort required to align a division of labor so that collective work gets done. In CSCW, this vocabulary describes the invisible coordination labor that sustains cooperative systems [46, 54]. Deployed AI-marketed K–12 systems remain comparatively under-examined through this frame, especially in accounts that follow classroom breakdowns into vendor-side repair channels.

2.3 Repair Studies

Graham and Thrift’s [19] “Out of order” and Jackson’s [29] “Rethinking Repair” jointly shift attention from breakdown as exception to repair as ongoing infrastructural achievement and skilled knowing. Functioning systems depend on repair labor that is often unacknowledged and de-valued [12, 51]. Mattern’s [34] work on maintenance and care, Denis and Pontille’s [15] study of material upkeep, and HCI work on repair politics and breakdown [28, 43] extend this line. Educational technology has received less repair-theoretic treatment; older work on computerized classrooms [11] does not address contemporary AI-marketed K–12 platforms in China.

2.4 Critical EdTech and the Politics of Educational Data Infrastructures

A fourth body of work, critical EdTech, treats educational technologies as commercial products, data infrastructures, and policy instruments—a position recently consolidated by Decuyppere, Grimaldi, and Landri [14] as a critical platform-studies stance toward education. Selwyn [47] argues that adoption vocabulary imports commercial assumptions into schooling; Williamson and colleagues [60–62] examine how educational data infrastructures structure what can be known about students and teachers, including how the pandemic accelerated platform-based datafication; Gulson, Sellar, and Webb [22] extend this to algorithmic governance of education policy; and recent work extends the

critique to platformized schooling and the techno-financial logics that fund it [25, 48, 63]. **Closest to the present argument** is Perrotta et al.'s [39] *distributed labour of platform pedagogies*: a frame naming how platforms route teaching work across human and algorithmic actors. In the Chinese context, Knox [31] analyzes how AI-in-education policy and private-sector enterprise interact, while recent work on Double Reduction situates education companies inside a broader governance project [3]. This paper extends such macro-level accounts downward, into the everyday repair labor that makes deployments operational, and adds vendor-attention asymmetry as a structural axis the distributed-labour frame does not yet emphasize.

2.5 Gap Statement

Together, these conversations show why adoption metrics are insufficient, why infrastructure and repair provide a better analytic vocabulary, and why commercial EdTech must be read as political economy as well as interface design. In one Chinese K-12 vendor's multi-module deployment, this paper contributes an empirical account of teacher and staff repair labor, the concept of visibility asymmetry across vendor-attention tiers, and a critical reading of "AI" / "smart" labels.

3 Setting, Positionality, and Methods

3.1 Research Setting and Technical Disclosure

The Deployed Suite. The field site is one Chinese educational-technology vendor (henceforth "the Vendor") that deploys a K-12 educational-technology suite across multiple regions. Three modules are relevant:

- **Module-C** ("the classroom-interaction system"): used during instruction and pre-class preparation. Functions include: lesson-courseware authoring and delivery, electronic textbook display, in-class polling and "all-class answer," random student selection, points tracking, screen-pushing between teacher and student tablets, and several subject-specific tools. Deployed across primary and secondary schools, with hardware ranging from teacher-tablet-only configurations to full deployments with student tablets. In several sites, Module-C deployment also depended on on-site staff for teacher training, tablet preparation, classroom setup, and first-line troubleshooting.
- **Module-H** ("the homework-workflow platform"): used post-class for assignment design and review. Functions include: scanning paper assignments via tabletop scanners, OCR-based recognition of student responses, answer matching for objective and constrained fill-in items, per-question correctness analytics, recommendation of follow-up questions keyed to knowledge-point tagging, layout and printing of answer sheets, and generation of per-student "error booklets" after each exam. Deployed primarily at high schools preparing for the *gaokao* (college entrance examination), with a delivery model that places vendor staff on-site to handle printing, distribution, and scanning.
- **Module-A** ("the AI Innovation Education module"): a programming-and-robotics module deployed primarily as after-school enrichment in primary schools. Components include programmable robot kits, a block-based programming environment, and a platform exposing several AI model APIs. I did not focus on Module-A in this analysis but include one teacher voice (T09) as a boundary case for the visibility-asymmetry and "AI" labeling discussions.

The three modules serve overlapping teacher and student populations, but do not share a unified sales channel, staff team, or backend dashboard. I treat them as **three deployment lines of a single Vendor** that intersect institutionally and in classrooms rather than as one technical platform.

The Vendor reaches schools through institutional procurement; teachers are mandated end-users. I classify schools along one analytic axis: **the level of vendor attention each site receives**. This

is not an official Vendor category or a fully independent quantitative measure. It is constructed from the pseudonym key, deployment records, field notes, and participant accounts, using four interpretive indicators: product-side visits, stability of on-site staff coverage, demo / customer-reference status, and whether teacher concerns are routinely routed beyond the school and staff layer. Because the corpus does not measure each indicator systematically at every school, I use the tiers as a comparative device. The three tiers are:

- **Tier 1 — flagship deployment sites** (high attention): stable product-side attention, stable on-site staff, and demo / customer-reference status.
- **Tier 2 — standard-staffed deployment sites** (medium attention): assigned staff but less routine product-side attention.
- **Tier 3 — thinly-covered deployment sites** (low attention): least direct product-side attention, stretched staff coverage, and partial or full withdrawal from the Vendor workflow beneath compliance-level deployment.

The three tiers do not all derive from the same source. **Tier 1’s “flagship” label** is anchored in the Vendor’s own commercial vocabulary — the terms “flagship,” “demonstration site,” and “sample school” circulate internally as ways of naming sites with conversion potential or reference value (see also §4.5, Pattern 2). **Tier 2 and Tier 3** are researcher-constructed contrasts, defined by the four interpretive indicators above to mark sites that lack the visibility characteristics of Tier 1. This emic/etic split is itself analytically relevant: the Vendor’s deployment language is sharper for sites it intends to surface than for sites where breakdowns are more often absorbed locally.

At the school level, the sample contains four Tier-1 schools, three Tier-2 schools, and one Tier-3 school. At the interview level, Tier 1 includes five teacher or teacher-group records (T01, T03, T06, T07, T09), Tier 2 includes four teacher / administrator records (T02, T04, T05, T08), and Tier 3 includes one teacher record (T10). The only staff interview (S01) crosses tiers because the same on-site staff member covered School-S7 (Tier 1) and School-S8 (Tier 3). Because Tier 3 is thin, the visibility-asymmetry argument rests primarily on the Tier-1 vs Tier-2 contrast within Region R1—demonstration middle schools S5/S6 and standard public middle schools S2/S3, all deploying Module-C—with T10/S8 as a limit case.

What is “AI” in the Deployed Stack: A Technical Disclosure. The Vendor markets the suite as “AI-augmented teaching,” “intelligent classrooms,” and “AI-powered grading.” I use a more careful vocabulary because the rhetorical work of these labels is itself an object of analysis (see §5.3). The components visible in classroom use comprised:

- **OCR-based scanning** (Module-H) of paper assignments. OCR did not deliver the autonomous grading the “AI-powered grading” framing implies; recognition was higher for objective items and notably lower for open free-response (60–70% per teacher report⁴).
- **Answer matching for objective and constrained non-choice items** (Module-H): in classroom-visible use, OCR plus answer-key or semantic matching for constrained responses; open free-response still required teacher verification.
- **Knowledge-point-tagged question recommendation** (Module-H): tagged-bank retrieval weighted by per-student error rates—a classical retrieval-and-rule-based recommender, not an end-to-end learned model.
- **Per-question and per-student analytics dashboards** (Module-H, Module-C): standard descriptive statistics in tabular and graphical form.

⁴TeacherT07 quoting an on-site colleague, Module-H teacher interview, 13:06.

- **Speech recognition with pronunciation scoring** (Module-C): the more clearly machine-learning-based component, processing student audio through speech-recognition and pronunciation-evaluation models.
- **An AI-model-API panel** (Module-A) exposing several model APIs—text classification, face recognition with style filters, automated classical-poetry generation, voice synthesis—as the AI core of the Innovation Education curriculum.

To preserve technical precision, the following vocabulary is used throughout the paper: “**Algorithmic**” for rule-based or classical retrieval recommendation; “**OCR-based**” for scanning-and-recognition components; “**Smart**” and “**AI-augmented**” in scare quotes when reporting Vendor or industry framings.

Module names and functional descriptors throughout this section are deliberately abstract to limit re-identification; specific feature combinations have been generalized, and concrete product names, region and city identifiers, and competitor brand names are anonymized throughout the paper.

3.2 Positionality

The author was employed by the Vendor in a product-side role during the period from which the primary field materials were generated. The author conducted site visits, interviewed teachers, observed classroom sessions, and produced field diaries and site-visit summaries as part of standard internal work. The data were originally collected for product improvement, not academic research; this paper is a **retrospective interpretive analysis** conducted after the author’s departure from the Vendor and without a live research relationship to the firm.

This positionality affords access depth and structural familiarity: teachers spoke under the expectation that concerns might be routed to product teams; the author observed delivery runs and internal problem-triage; and the role provided a partial technical view of dashboards, adoption metrics, and roadmap response patterns. It also constrains the claims.

The data were not generated with academic analysis in mind; the corpus was not produced under a prospective academic human-subjects protocol; sampling reflects product field visits rather than theoretical sampling; and the fieldworker’s product-side role shaped what was said and observed. Following ethnographic precedents for the analysis of materials produced in one’s own prior workplace [17, 36, 38, 56], I treat the corpus as **field artifacts from an industry employment period** rather than formal ethnography. The full ethical reasoning that warrants this retrospective analysis without participant re-engagement is developed in §3.5. I do not make causal claims, and I anonymize individuals, schools, regions, competitors, and organizational details.

The author’s prior-employee status creates a structural asymmetry between the analytical frame and the participants’ own framings. Three mitigations were applied: (a) preserving teacher and staff voices through direct quotation where consistent with anonymization; (b) grounding interpretive claims in specific data points with explicit source markers (interview / observation / field note) so that the inferential distance from quotation to claim remains auditable; and (c) refusing analytical framings—for instance, claims about Vendor commercial intent or about teacher pedagogical adequacy—that would have been unavailable from within the participants’ own accounts. These mitigations bound the analysis without claiming to neutralize the positionality.

3.3 Corpus

The corpus draws from eight focal schools across three regions, summarized in Table 1; three data streams—detailed below—populate this site map.

Table 1. Site and participant map. PS = primary, MS = middle, HS = high school. “Records” lists interview IDs (T-prefix) and classroom observation IDs (Obs-prefix); see Tables 2 and 3 for participant durations and observed lesson topics.

Region	School	Tier	Level	Module(s)	Records
R1	S1	1	HS	Module-H	T06, T07
R1	S5	1	MS	Module-C	T01; Obs-5 (G7)
R1	S6	1	MS	Module-C	T03; Obs-1 (G8), Obs-2 (G9)
R1	S2	2	MS	Module-C	T02
R1	S3	2	MS	Module-C	T04
R2	S7	1	PS	Module-C, Module-A	T09; Obs-3 (G6), Obs-4 (G4)
R2	S8	3	PS	Module-C [†]	T10
R3	S4	2	HS	Module-H	T05, T08

[†] system-withdrawn deployment. S01 (on-site staff) covers S7 (Tier 1) and S8 (Tier 3), cross-tier.

Interviews and brief elicitation records (n=11). Semi-structured conversations conducted on-site during the employment period. **Ten teacher or administrator records** span the three modules of the suite; two very short records are treated as brief elicitation rather than full interviews. **One on-site delivery staff member (S01)** provides the staff-side voice across two schools; due to ethical considerations described in §3.5, S01’s biographical details are generalized and direct quotation has been replaced with author paraphrase.

Table 2. Interview and elicitation records.

ID	Role	Module	Grade / Subject	School	Tier	Duration
T01	Classroom teacher	Module-C	G9 physics	S5	1	~00:17:00
T02	Classroom teacher	Module-C	G8 English	S2	2	00:23:28
T03	Teacher + students (group)	Module-C	G8 math (post-class)	S6	1	00:06:40
T04	Classroom teacher	Module-C	G9 English	S3	2	00:58:21
T05	Vice principal + teacher	Module-H	G10 math	S4	2	00:04:52
T06	Classroom teacher	Module-H	G12 math	S1	1	00:27:54
T07	Classroom teacher	Module-H	G12 chemistry	S1	1	00:21:27
T08	Subject-group head	Module-H	G12 math	S4	2	00:19:11
T09	Classroom teacher	Module-A	G5–6 AI Innovation Ed	S7	1	00:32:46
T10	Classroom teacher (system-withdrawn)	Module-C	G4 math	S8	3	00:44:57
S01	On-site delivery staff	Module-C	Cross-site staff account	—	1, 3	01:07:01

Classroom observations (n=5). Five in-classroom observation sessions conducted during the employment period, all covering Module-C in deployment (Table 3). Each observation produced a structured record covering teaching segments and timing, instructional activity description, product features used, and observed problems / friction.

Table 3. Classroom observations.

#	School	Grade / Subject	Lesson topic	Findings supported
Obs-1	S6	G8 math	piecewise functions	§4.1 in-class friction
Obs-2	S6	G9 history	exploration of new sea routes	§4.1 in-class friction; §4.5 cross-tier random-selection
Obs-3	S7	G6 writing	“a writing journey”	§4.5 cross-tier random-selection
Obs-4	S7	G4 Chinese	unit review	§4.3 Resource output; §4.1 pre-class friction
Obs-5	S5	G7 geography	longitude and latitude exercises	§4.1 pre-class setup; §4.3 Layout output

Documentary artifacts. 28+ days of field notes from eight focal schools and related site visits across four regions. These notes include site-visit records, classroom-observation notes, analytic memos, and deployment-context notes. I use them as contextual and triangulating artifacts rather than as a complete audit trail of every product issue.

3.4 Analytical Approach

I conducted an interpretive thematic reading of the corpus, drawing on reflexive thematic analysis [5, 6]. The reading was both top-down, using sensitizing concepts from infrastructure and repair studies, and bottom-up, allowing local patterns to refine themes. Interview materials included recordings, notes, and excerpt-level summaries; classroom observations and field notes were written during or shortly after site visits. Chinese excerpts were translated by the author, with originals retained only where analytic value outweighed residual anonymization risk.

Coding was operationalized as **50 excerpt-level analytic memos** maintained in Obsidian. Each memo captures one analytic episode and carries structured frontmatter (source, school, tier, module, labor form, evidence strength, triangulating memos), followed by the excerpt, context, and an analytic note linking the episode to a specific claim. Tagging follows a controlled namespace over labor form, site, tier, module, role, and evidence type; the memos are indexed by source, labor form, and tier. An anonymized example memo (T06-01) is available as supplemental material where permitted.

These memos are **retroactive** relative to the paper's first analytic pass: they systematize the evidence base after the labor-form vocabulary had crystallized, placing the analysis on the deductive end of reflexive thematic analysis [6] rather than within a prospective open-then-axial pipeline. What the memo set provides is **auditability**: each claim in §4–§5 is grounded in a named excerpt, with explicit triangulation links where corpus-internal corroboration exists and an evidence-strength rating that flags single-stream claims distinctly from cross-stream triangulated ones.

The analytical process proceeded by extracting episodes of work required to make the system function, memoing them, grouping into candidate themes, and checking themes across data streams; divergences were retained as theoretical content—for instance, the gap between what dashboards record as “active interaction usage” and what the on-site staff account reports about performative usage patterns (§4.5) is itself a finding, not an inconsistency to be resolved.

The analytical structure spans three data streams:

- **Articulated** (interviews) — what teachers and staff say they do and why.
- **Observed** (classroom observations) — what teachers do in situ during instruction.
- **Documented** (field notes and site-visit summaries) — what gets recorded as problem at the institutional / product layer.

3.5 Ethics

The corpus was generated during the author's employment at the Vendor as part of standard product-fieldwork practice. Teachers and on-site staff gave verbal agreement to record interviews and to use field-visit notes for internal product-improvement purposes; **the present academic analysis falls outside that original consent scope.**

Three considerations shaped that reasoning. First, **re-engagement after the author's departure would reactivate the asymmetric vendor-school-teacher relationship** under which original participation occurred: requests routed through former colleagues or school administrators would carry institutional pressure participants did not assume for academic publication, and could plausibly produce coerced rather than informed consent. Second, the analysis makes **no individual-level claims**—about teacher competence, school performance, intent, or causal effect—that would require the kind of consent re-engagement could legitimately establish; the claims are structural and mechanism-level, calibrated to what corpus-internal triangulation can support (§3.4). Third, the paper **reduces what is disclosed to the minimum required to ground the structural argument**, applying layered pseudonymization, paraphrase of high-risk staff passages, and refusal of identifying detail tuples.

The ethical framework draws on the AoIR *Internet Research Ethics* guidance for non-consensual data reanalysis [32], the AERA *Code of Ethics* for educational research with vulnerable populations [1], and ethnographic precedents for retrospective analysis of one’s own prior workplace [17, 36]. The Menlo Report [30] informed disclosure-minimization choices as a secondary framework, given its origin in ICT security rather than workplace or educational settings. I present this as a practitioner-grounded interpretive account rather than conventional ethnography.

Anonymization alone does not neutralize re-identification risk in a corpus where module, tier, region, school education-level, and staff role co-vary. A reader inside the Vendor with access to internal records could in principle recover specific schools or participants from these combinations. I have therefore taken layered steps to reduce this surface: I anonymize at region, city, district, and school; I abstract module names to functional descriptors (Module-C / H / A); I generalize the biographical detail of the single on-site staff participant (S01) and paraphrase her account in place of direct quotation; and I avoid tuples of detail (region + education level + module + tier + grade + role) that would narrow identification. I also avoid claims the corpus cannot support. The largest residual risk attaches to S01 and to the Tier-3 limit case (School-S8); the strongest mitigations are concentrated there.

I do not re-contact participants through the former vendor relationship. Declining re-contact is an ethical choice, not a convenience: the corpus’s analytic ceiling is set lower in exchange for not extending the original consent gap. The author retains no employment relationship, no access to Vendor systems, and no live data feed; the manuscript discloses only what is necessary to ground the argument, without reproducing internal vendor documents. Remaining limits — that vendor field artifacts cannot be converted into prospective ethnographic data, that participants have no academic-context withdrawal mechanism, and that student and parent perspectives are absent — I return to in §5.4.

4 Findings

I present four repair practices: **temporal stitching** (§4.1), **defensive redundancy** (§4.2), **content re-verification** (§4.3), and **mediation labor** (§4.4), followed by a cross-cutting analysis of how repair becomes unevenly visible across **vendor-attention tiers** (§4.5).

4.1 Temporal Stitching: Reconciling Multi-Layer Time Mismatch

The deployed suite is designed around a specific assumption: that classroom and homework workflows have time budgets flexible enough to absorb the platform’s operational tempos. The corpus shows this assumption breaking at three temporal layers—**within-class** transactions where minute-scale frictions accumulate; **day-cycle** loops where homework workflows fail to fit Chinese teaching’s same-day iteration requirement; and **week-and-cycle** pressure where curriculum density and assessment cadence saturate institutional time budgets. Repair labor accumulates at each of these joints.

In-class friction. Five classroom observations document that platform feature use carries substantial concealed time costs absorbed during instruction.

Pre-class setup itself is non-trivial.⁵ In Obs-5, the teacher and an on-site staff member spent **15 minutes** distributing student tablets, labeling devices, managing login, and troubleshooting failures—time invisible in Module-C’s dashboards, which begin counting “interaction events” only after class begins. T04 reports a cognate pattern: vendor cloud sync between her personal computer

⁵TeacherT02 reports an analogous 5-minute tablet-preparation burden per cycle.

and the classroom tablet fails routinely, costing 2–3 minutes of re-download and re-upload labor before each class.⁶

Mid-class, interaction features themselves accumulate friction. In Obs-1, the teacher attempted to switch interaction modes mid-lesson; the switch required ending the prior session—which deleted its data—then opening a new one, during which the program crashed and could not be re-entered.⁷ The teacher reported afterwards that she had taught the same lesson three times before she could fit the interaction-rich version into a single class period. TeacherT01 reports a quantified version of the same delta: a calculation problem demonstrated on the board takes 10 minutes; the same problem routed through photo-upload, share, and explain interaction takes 15 minutes.⁸

At the limit, friction produces real-time feature abandonment. In Obs-2, random-selection picks failed four times in a row; the teacher dropped the feature mid-lesson and reverted to direct explanation, attributing the difficulty afterward to G9 curriculum density—a pattern T01 articulated independently.⁹

These small frictions cumulatively constitute a daily time cost that teachers absorb through compressing other lesson elements, dropping interaction features mid-class, or pre-rehearsing at home to ensure platform-failures do not break in-class flow.

Daily transaction cycle. TeacherT06 describes the daily Module-H workflow in concentrated form. She teaches through the morning; afternoons are nominally available for **same-day** assignment preparation. “You cannot know in the afternoon what you’ll need to assign today”¹⁰ until the morning’s class has happened. Same-day iteration is pedagogically required, but the platform’s four-hour print-and-deliver service level makes it **very difficult**. TeacherT06 has negotiated with on-site delivery staff for an emergency turnaround; even so, she has only an afternoon window in which to pick items, hand off to printing, and absorb residual mismatch as personal labor.¹¹ Another related tempo failure surfaces in question-bank update cycles: she uploads newly circulating mock-exam papers herself before the platform adds them to the official bank.¹² Teachers compensate via lateral peer networks and by manually ingesting PDFs into the scanning workflow.

Curriculum and cycle pressure. The day-cycle gap is reinforced by week- and multi-week-scale pressures the platform’s design does not anticipate. TeacherT05 (vice principal, S4) describes the underlying structural pressure: “now that we have *double-reduction* and two-day weekend, class hours themselves are tight; once a class ends, you cannot just teach this material—you have to teach the next core lesson.” Time available for any algorithm-recommended supplementary work is structurally squeezed by the mandated curriculum sequence and by the policy compression of school days, consistent with recent analyses of Double Reduction as a broad reorganization of school, family, and tutoring responsibilities rather than a narrow tutoring-market intervention [3]. TeacherT07 (chemistry, S1) reports a parallel constraint at the assessment-cycle level: “our exam frequency is very high now...about a round every two weeks across subjects...if [the platform] adds per-student error booklets on top of that, that’s basically two extra packets per cycle.” Together with TeacherT06’s daily-cycle gap above, these three voices document temporal mismatch at three distinct scales the platform’s design does not anticipate (see Table 2). Repair at this layer is institutional rather than personal: teachers with formal authority (subject-group heads, vice

⁶TeacherT04, Module-C teacher interview, 26:37.

⁷Observation 1.

⁸TeacherT01, Module-C teacher interview, 07:42–08:22.

⁹Observation 2.

¹⁰TeacherT06, Module-H teacher interview, 23:57–24:02.

¹¹TeacherT06 and on-site staff in same interview, 24:05–24:27.

¹²TeacherT06, same interview, 03:22.

principals) carve out designated time slots for system use within otherwise saturated schedules—a form of repair labor discussed in §4.4.

In adoption dashboards, this temporal-stitching labor registers only as regular Module-H use or Module-C interaction use; its structure, and the tempo mismatch that produces it, remain invisible.

4.2 Defensive Redundancy: Maintaining External Backups Against System Unreliability

The deployed suite is designed around a second specific assumption: that its information substrate—what it retains, retrieves, and represents—is reliable enough that teachers and on-site staff can hand off to it without preparing for failure. The corpus shows this assumption breaking at three reliability layers documented below. Teachers and staff respond by maintaining **external, materially redundant records and infrastructures**—paper copies, USB drives, parallel non-Vendor workflows, daily manual hardware preparation—that they can count on because they themselves produced and maintain them. I call this pattern **defensive redundancy**: pre-emptive labor invested before failures manifest. It differs from temporal stitching (§4.1) in being primarily about supplementing the system’s representational and operational fidelity rather than reconciling time.

State redundancy. TeacherT06 reported a recurring state-retrieval failure: teachers could not reliably locate assignments they had published themselves. “Sometimes I find that the assignment I published—I can’t find it myself.”¹³ Field notes from the same visit treated this as a workflow problem rather than memory lapse: the platform’s “my published assignments” view sits inside a multi-step navigation path, and published answer keys are packaged inside an archive whose button text does not indicate that answers are included. TeacherT06’s response is anticipatory: she requests extra paper copies at print-time, **before** she knows she will need them, recovering answer keys later from students’ physical sheets. The paper layer is redundant infrastructure she trusts because she controls it.¹⁴

Hardware redundancy. S01 described the routine in which she and her colleagues spent extended periods winding cables, plugging tablets into an unreliable charging dock, and arranging them in correct orientation. T02 describes the school-side mirror of this pressure: shared tablets are hard to manage, students forget to charge them, and assigning one student to take responsibility creates its own problem.¹⁵ The teacher cannot reasonably absorb the labor, cannot reasonably delegate it, and yet bears the downstream cost when classroom activity breaks. Hardware reliability falls into an unowned gap between staff members’ daily cabling and the teacher’s mid-class breakdown.

System redundancy. Parallel infrastructure exists across the corpus as a spectrum of scope. S01 describes routinely running both the Vendor system and a competitor system in parallel windows during instruction—the only condition some schools would tolerate.¹⁶ At Tier 2, T04 migrated reading-task assignment to a third-party tool when the Vendor’s feature failed for back-row students, while keeping other Vendor channels.¹⁷ At the Tier-3 limit, T10 maintains lesson preparation, supplementary materials, and in-class tools entirely outside the Vendor suite, logging in only when administrative compliance requires it. Across this spectrum, redundancy escalates from tolerated co-existence (S01) to channel-specific substitution (T04) to comprehensive substitution (T10). At the limit, the deployed system has effectively ceased to function as infrastructure and has been demoted to a compliance interface.

¹³TeacherT06, Module-H teacher interview, 16:44.

¹⁴State-retention failures take other forms in the corpus: T02 reports progressive save corruption after repeated edits; T04 reports data loss in the points-system across semesters before backend rollback.

¹⁵TeacherT02, Module-C teacher interview, 11:43–11:58.

¹⁶On-site staff interview.

¹⁷TeacherT04, Module-C teacher interview, 14:36.

Across the three layers, defensive redundancy delivers **trust under conditions of unreliable infrastructure**. The dashboard registers none of this: extra paper copies disappear into ordinary use data, and a teacher with a complete parallel workflow appears merely as a low-frequency user. The teacher-borne labor of system-side reliability is invisible.

4.3 Content Re-verification: The Boundary of Algorithmic Substitution

The deployed suite assumes its algorithmic content outputs can stand in for the contextual judgment connecting the teacher–student–exam triad. This assumption is **structurally incomplete**: even calibrated recommendation, accurate OCR, and curated resources would not eliminate the need to gauge this student’s stuck points, this class’s pace, this week’s exam horizon, or this teacher’s pedagogical priorities. Teachers respond by checking, correcting, or replacing platform-produced content before it reaches students—a pattern I call **content re-verification**: cognitive labor that supplies the contextual judgment the platform omits across recommendation, grading, layout, and resources. It differs from defensive redundancy (§4.2) in supplying contextual judgment rather than backstopping operational reliability.

Recommendation output. The Module-H recommendation feature fails not at any single calibration parameter but at the level of its design premise. Two interview voices document compensatory labor at two distinct points along this premise’s failure surface—at the level of **bank fit** (T06), eliciting **active circumvention**, and at the level of **difficulty calibration** (T07), eliciting **passive absorption** of the failure cost.

Bank-fit failure. TeacherT06 (Module-H, S1) put it directly: “the questions it pushes are not necessarily the ones I need.” She clarified that this was not a quality complaint—the algorithm-recommended items were not obviously poor—but a **fit** complaint: the platform draws from a different question bank than the legacy one her subject-group has curated. Her response was to abandon the recommendation path entirely and use the platform as a print-pipeline on top of the legacy bank—a use pattern the dashboard registers as “engagement” but that bypasses the algorithm.

Difficulty-calibration failure. TeacherT07 reported that algorithm-generated booklets, in a context where students could not actually do the items, end up unused—“they stack up...in their desk drawers.”¹⁸ T05 echoed the underlying mechanism: “it pushes too difficult; the students literally can’t do them.”¹⁹ TeacherT08 added the design-level diagnosis: common and individualized items bundled together too coarsely, so students set the booklet aside.²⁰ T07’s structural fix would widen the difficulty spread; absent that, her repair is manual item-substitution before assignment.

What unites these two voices is not the calibration parameter that fails but the polarity of the compensatory labor they generate—active circumvention at one end, passive absorption at the other. In both cases, the teacher’s judgment about *which items, at what difficulty, for what students* is doing the work that the recommendation feature claims to do. The dashboard registers this as “use of the recommendation feature”—but the algorithm’s contribution has been reduced to a print pipeline running on teacher judgment.

Grading output. Grading is the channel where the system delivers real labor savings: without OCR, T06 would hand-grade objective items as well. The repair argument operates one level up. Classification—matching answers against ground truth—is one task; grading as a pedagogical act is another, requiring partial-credit decisions, diagnosis of what wrong answers reveal, and judgment about which errors warrant re-teaching. T07 reports using the system for “the overall data” and

¹⁸TeacherT07, Module-H teacher interview, 02:02, 03:30.

¹⁹TeacherT05, Module-H teacher interview, 03:55.

²⁰TeacherT08, department-head interview, 08:13–13:48.

hand-grading items herself.²¹ T06 runs the same pattern: she treats the per-question analytics as a scoreboard and does the rest manually. The dashboard collapses classification and grading judgment into one credit; the latter remains uncounted.

Layout output. T06 reformats the auto-generated paper layout every assignment cycle (three-column, A3, row-height, font adjustments) until it fits her pedagogical context. Obs-5 documents a digital-layout cognate at School-S5: the per-question correctness page required manual alignment of question numbers and uploaded images with the printed worksheet; without that, students were left switching between screen and paper during review. In both registers, the auto-generated layout is technically produced; its pedagogical fit is re-fitted by the teacher.

Resource output. The Vendor’s courseware library claims to substitute for the teacher’s curation labor. The claim holds at the **form layer**—the library indexes, searches, and renders content; it admits or excludes file types and third-party tools. It does not hold at the **substance layer**: classroom-ready content requires alignment with this class’s reading level, this week’s curriculum focus, this subject group’s existing materials, this teacher’s presentation style. Substance failures appear along two axes.

Content quality and depth. The on-site staff member relayed teachers’ framing: competing courseware could be used directly, while the Vendor’s materials had to be pieced together.²² Her own assessment is that the Vendor library is largely aggregated from external download sources rather than originally produced. T10 echoes the user-side: the resources are “too many and too messy... even teachers can’t tell them apart.”²³ At the milder end, Obs-4 observed embedded video and image resources displayed at low resolution. At the severe end, T10 demonstrated a music courseware retrieved from the library that contained no audio at all and only one partially usable item.²⁴ Across the gradient, the failure is one kind: the library’s substantive content is shallower than classroom use requires.

Subject-tool admission. Even where library content is acceptable, the platform’s form layer admits some substantive subject tools and excludes others. T03 names this for mathematics: a third-party dynamic-geometry tool central to her pedagogy cannot run on the student tablets and is absent from the platform’s application store. Her courseware preparation has shifted to a home desktop; she exports static slides into the deployed system rather than letting students manipulate dynamic figures.²⁵ The substance her subject demands lives in tools the deployed form layer does not admit.

Repair labor sits where the form layer ends and the substance layer begins. Teachers maintain parallel libraries—local files on USB drives, courseware authored in a competitor’s platform, materials downloaded from external resource sites—and pre-screen the platform’s offerings rather than relying on its recommendations. Where library content is shallow enough that pre-screening yields nothing to work with, repair extends to sourcing or building substance from elsewhere.

A subsidiary finding: teachers and staff endorse the design premise but question its substitutability. TeacherT07 was explicit: “the concept is good, but the quantity is too much, and students cannot use it.”²⁶ A brief field-note record corroborates this stratified pattern from the student side: in short exchanges with three students, higher-achieving students used the booklet selectively, while another reported not using it because the items were not actionable.²⁷ The on-site

²¹TeacherT07, Module-H teacher interview, 13:05–13:51.

²²On-site staff interview, 00:36–04:37.

²³TeacherT10, classroom teacher interview, 04:20–06:08.

²⁴TeacherT10, classroom teacher interview, 17:01–18:21.

²⁵TeacherT03, classroom teacher interview, 02:57–06:09.

²⁶TeacherT07, Module-H teacher interview, 02:02.

²⁷Field notes, late 2024. The three students were approached for short conversations during a school visit, not systematically sampled; I treat this as anchoring rather than representative evidence.

staff member characterized many marquee features as “icing on the cake” relative to teachers’ actual demand for usable resources. The pattern is consistent across all four content channels: teachers and staff are continuously investing cognitive and practical labor into the system, and this labor makes the system’s outputs classroom-ready. Re-verification is the labor that occurs at this boundary.

4.4 Mediation Labor: The Multi-Actor Path Behind Platform Adoption

The deployed suite also assumes adoption is something the teacher does at the platform interface. The vendor’s deployment apparatus tacitly acknowledges otherwise (through on-site staff, school administrators, and feedback channels), yet adoption metrics still treat the teacher’s click as the unit of credit and analysis. The corpus shows mediation across within-school actors (subject-group heads, vice principals, classroom teachers as relays to students) and vendor-side actors (on-site staff, school administrators). I call this **mediation labor**: the cognitive and relational work—translation, routing, and access procurement—that bridges layers the dashboard treats as direct. It differs from content re-verification (§4.3) in routing across actors rather than supplying contextual content.

Classroom teachers sit at the base of every mediation chain. They absorb the first-order costs of system misalignment: pre-class prep time, lost-assignment backup, item-by-item quality checks, and in-class workarounds when interaction features fail. Their own mediation moves in two directions: instruction relays each system-mandated task to students (without that relay, no platform output reaches classroom activity), and recurring breakdowns escalate only when teachers file requests through on-site delivery staff. After repeatedly losing track of her own published assignments, TeacherT06 filed a request through on-site staff for a “my recent assignments” view; field notes record that this request was logged at the staff level but had not reached the product roadmap by the end of the survey period.

Within-school mediators above the classroom—subject-group heads and vice principals—route concerns at registers the classroom level cannot reach. TeacherT08, a subject-group head at School-S4, performs **boundary-spanning labor**: she translates between operational scales (the classroom teacher’s daily concerns and the platform’s design logic) and organizational scales (the teaching staff under her and the school administration above). Her account shows two movements. First, from operational concern to platform-level design proposal: her suggestion to separate shared and individual error items into distinct print bundles is articulated as design recommendation, not workaround—asking the platform to redesign, not asking the platform-as-given to be tolerated. Second, intra-group labor coordination: a designated showcase teacher in the math department picks items for the entire group, converting what would otherwise be eight teachers’ question-picking labor into one teacher’s labor for eight.²⁸ TeacherT05, vice principal at School-S4, performs mediation at a register the platform’s design entirely fails to anticipate: **institutional time allocation**. Module-H assumes a student time-budget in which algorithm-recommended supplementary content has room to land; T05’s school does not have such a budget—students’ time is saturated by standardized prep materials, external mock exams, and the school’s own assessment cycle. T05’s mediation is to **carve out institutional time**: a biweekly “push session” for her math classes under administrative coordination. “My teachers don’t have time to help students with this; we are lucky to get through the day’s lesson content. So if I don’t push, nothing will happen.”²⁹ A cognate appears at the department-head level: T08 reports a designated

²⁸A cognate but differently routed pattern appears at School-S1: T07 describes a math-department colleague whose item-picking labor serves multiple classes, activated through ongoing interpersonal pressure rather than administrative designation.

²⁹TeacherT05, Module-H teacher interview, 00:00.

“error-review time” on Friday and Sunday evenings, an institutional slot pre-existing the platform into which Module-H output is fitted rather than the reverse.³⁰

On-site delivery staff—vendor employees embedded at partner schools—perform the most extensive and least visible category of mediation. Drawing on the staff interview and field-note observations of delivery runs, the staff role spans:

Translation between vendor and teacher. Translation is **curatorial labor** that runs in two directions. In the forward direction, the on-site staff member trains teachers and decides which features to push; she characterized marquee features as “icing on the cake” relative to teachers’ actual demand for usable resources.³¹ In the reverse direction, she collects teacher-side concerns and renders them into vendor-actionable formulations during product-manager visits or field-input requests.³² In both directions translation is selective.

Indirect engagement: admin routing and cycle interventions. Indirect engagement is **structural labor**: working the relationships and timing that produce teacher attention in the first place. Direct engagement with teachers, who are too overwhelmed by their own teaching load to volunteer time, fails. The staff member routes through administrative channels, making platform engagement contingent on ongoing relational labor with school administrators. The same routing form appears in the booklet weekly cycle: field notes record a four-actor task matrix in which the on-site staff member coordinates rollout across teachers, subject-group heads, and administrators.³³ Within the cycle, her interventions are timed: after each exam, she proactively meets with teachers to coordinate booklet assignment, especially before weekends.³⁴

Roles are not equal: routing produces selective visibility. What links the four roles described above is not a uniformly distributed mediation burden but a routing structure that varies by position. Within the school, capacity to make concerns visible upward increases with formal position—classroom teachers absorb first-order breakdowns; subject-group heads formulate design proposals (T08); vice principals carve out institutional time (T05)—but every role is bounded by the same *gaokao*-and-curriculum pressure. Higher position gives concerns more reach; it does not remove the classroom conditions that generate the labor.

On the vendor side, the on-site staff member occupies a bilateral-broker position—accountable upward to vendor KPIs and laterally to the school relationships that sustain her presence. One route to the product team runs through her, and that route is mediated, selective, and capacity-constrained, as documented above.

The consequence is not absolute blockage but **selective visibility**. The dashboard counts a teacher’s interaction event; it does not count the staff work that made the feature available, the administrator work that induced its use, or the staff translation that turned an observed problem into a possible product conversation. These are differences in structural positions of mediation work that the adoption frame and the dashboard together render invisible. The patterns through which this invisibility is unevenly distributed across vendor-attention tiers are developed in §4.5.

4.5 Visibility Asymmetry Across Vendor-Attention Tiers

A cross-cutting pattern runs through the corpus: similar breakdown categories appear at Tier-1 (flagship), Tier-2 (standard-staffed), and Tier-3 (thinly-covered) sites, but the **routing conditions**

³⁰TeacherT08, Module-H teacher interview, 06:54.

³¹On-site staff interview, 09:39–09:53.

³²Synthesized from staff interview. The reverse channel is mediated and capacity-constrained: what travels depends on staff bandwidth, teacher availability, administrator permission, product-line timing, and whether the issue can be framed in a form the product team recognizes.

³³Field notes from primary corpus period.

³⁴Field notes from primary corpus period.

Table 4. Cross-tier evidence for visibility asymmetry.

Breakdown	T1	T2	T3	Routing
Random selection / interaction	Obs-1; Obs-3; T01	T04	—	T1 observed or staff-routed; T2 teacher-routed
Tablet setup	Obs-5; S01	T02	S01	Off-dashboard across tiers
Low-quality courseware	Obs-4	—	T10	Flagship observable; limit case drives withdrawal
Homework recommendation / scoring mismatch	T06; T07	T05; T08	—	T1 surfaces through attention; T2 depends on role-based escalation
Compliance-oriented interaction	S01	—	S01	Counted as adoption despite performative use

under which a breakdown becomes product-visible vary with vendor attention. I call this **visibility asymmetry**. The asymmetry runs along the vendor-attention axis defined in §3.1, not along school education-level. Table 4 cross-cuts §4 by breakdown category, showing the corpus distribution of each across vendor-attention tiers and the routing conditions under which each becomes product-visible.

Definition. The four labor forms documented in §4.1–§4.4 recur across vendor-attention tiers, yet only some of that labor becomes legible at the product layer. I call this property *visibility asymmetry*: the structural feature of vendor-managed sociotechnical infrastructure under which similarly-formed classroom breakdowns reach product repair channels at unequal rates, conditioned on each site’s position in a vendor-attention hierarchy. The asymmetry is not in whether breakdowns occur but in whether the local repair labor that absorbs them surfaces as a product-side record or remains absorbed in teacher and staff workflows.

Mechanism. The mechanism unfolds across four coupled stages, each visible in the corpus (summarized in Figure 1):

- (1) *Procurement and conversion logic* stratify deployed sites into vendor-attention tiers — flagship, standard-staffed, thinly covered — according to commercial categories such as conversion potential, contract stage, and demonstration value (elaborated below).
- (2) *Differential staffing and visit cadence* track those tiers: stable on-site staff and frequent product-team visits at high-attention sites; episodic or absent presence at low-attention sites.
- (3) *Routing channels* convey local breakdowns into product-visible records only where staff or relational paths are in place; elsewhere, repair is absorbed locally into teacher and staff workflows.
- (4) *Dashboard and adoption metrics* encode the resulting distribution: high-attention sites surface as sites of legible repair; low-attention sites surface as sites of “adoption,” with the repair labor that produced that adoption invisible by construction.

Relation to prior work. Star [49] and Bowker and Star [4] establish that infrastructures become visible upon breakdown and that classification systems make some labor invisible; articulation-work scholarship [46, 52] describes the connective labor that holds cooperative systems together. Visibility asymmetry extends this tradition by coupling infrastructure-visibility analysis to a **vendor-side commercial attention economy**: in deployed commercial EdTech, which breakdowns become visible is conditioned not only by category structure but by the procurement architecture that decides which sites get attended to at all. The concept is therefore CSCW-shaped — about visibility jointly produced across cooperating actors and infrastructures — but it adds a commercial-routing layer that the original infrastructure-studies tradition did not need to theorize.

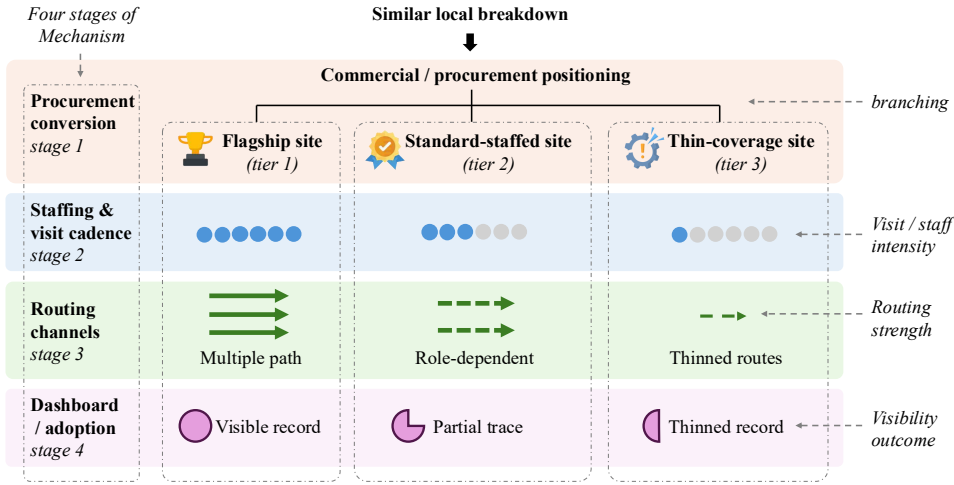


Fig. 1. **Visibility asymmetry as a four-stage routing mechanism (§4.5).** Across procurement, staffing/visits, routing, and dashboard stages, vendor attention conditions whether a similar local breakdown surfaces as a product-visible record, partial trace, or thinned record. Visual gradients are directional, not quantitative.

At Tier-1 (flagship) sites—School-S1 in Region R1, the demonstration middle schools School-S5 and School-S6 in the same region, and the demonstration primary school School-S7 in Region R2—breakdowns are not absent. They are more likely to be attached to a repair channel. Product-side visits and field observations occur more often; on-site staff are stably assigned; some teachers and subject leaders have more plausible routes for translating local frustration into product-facing claims. This is visible in the opening cases. TeacherT06’s published-assignment problem is not merely a personal workaround; it becomes a staff-mediated request for a “my recent assignments” view. At School-S7, the QR-code, low-resolution resource, random-selection, and tablet-holding problems occur inside a high-attention primary-school deployment. The S7 case therefore corrects a too-simple reading of visibility asymmetry: flagship status does not mean the system works without repair. It means the breakdown is more likely to be observed, narrated, and routed.

At Tier-2 (standard-staffed) sites—School-S2 and School-S3 in Region R1 for Module-C, and School-S4 in Region R3 for Module-H—the same classes of breakdowns surface, but the route upward is more episodic and role-dependent. Within Region R1 and Module-C, the corpus provides a directly matched comparison: Tier-1 middle schools S5 and S6 and Tier-2 middle schools S2 and S3 are in the same city and deploy the same module. Across those schools, tablet management, random-selection friction, and in-class interaction costs recur. T02 describes the unowned labor of keeping shared tablets charged; T04 describes random selection repeatedly selecting from a small visible subset, wasting class time; the corresponding observed Tier-1 cases at S5/S6 show similar classroom frictions under more product-visible conditions. In Module-H, School-S4’s T05 and T08 interviews show that standard-staffed sites can articulate sophisticated design needs, but those needs travel through administrator and subject-head standing rather than through a stable product channel.

At Tier-3 (thinly-covered) sites, the present corpus contains only School-S8. I therefore treat S8 as a limit case rather than as a basis for broad claims about all under-resourced schools. T10's account shows what defensive redundancy looks like when it expands into a complete parallel workflow: lesson preparation, resources, and interaction practices move outside the Vendor system, while the Vendor platform remains present mainly as a compliance surface. This is the most extreme form of local absorption in the corpus. It is analytically important, but it cannot carry the full evidentiary weight of the tier argument by itself.

Two patterns frame this asymmetry: a differential routing of attention across sites, sorted by a commercial-procurement architecture that assigns it. *Differential routing* transforms classroom articulations into different visibility states. At Tier-1 sites with stable visit cadence, articulations enter product-team field-visit records: TeacherT01's account of the random-selection algorithm's reverse-bias was logged during a Tier-1 site visit.³⁵ At Tier-2 sites without comparable cadence, the same articulation requires teacher persistence: TeacherT04 reports the points-system data-loss problem recurred four times before backend rollback, only after escalation through several vendor staff and repeated insistence that the problem was vendor-side.³⁶ Within a single staff caseload, S01 reports concentrated daily presence at one site while another received thinner coverage and no product-team visit for six to twelve months.³⁷ *This routing differential is sorted by commercial-procurement categories that vendor practice articulates.* Field notes record vendor-internal terminology naming the logic: "flagship sites" and "sample schools" are defined by conversion potential, "fan teachers" are cultivated through ongoing relational labor, and pricing is structured by deployment frequency.³⁸ Procurement architecture operates similarly: T10 reports that the bureau procures Module-C upgrades for all district schools simultaneously, not per-school.³⁹ The Tier-3 limit case shows what happens when commercial deprioritization meets this architecture: T10 reports active engagement with the vendor system had narrowed to two school actors.⁴⁰ As a single-case observation, S8 functions as the corpus's outer limit; commercial logic is not the sole driver—staff capacity, contract stage, and product-line timing also matter. The account is consistent with critical-EdTech analyses of educational data infrastructures [25, 47, 48, 60, 61]. Commercial vocabulary and bureau procurement architecture do not sit alongside the routing differential; they assign it.

Together, these two patterns characterize a structure in which repair labor accumulates unevenly: visibly routed at Tier-1, contingently at Tier-2, locally absorbed at the Tier-3 limit case. The corpus shows the reverse causal direction from common assumption: commercial categories assign sites to routing channels, and routing channels assign breakdowns to visibility states. This has the distinctive shape of what Star [49] calls the **political shape of infrastructure**: what counts as infrastructure—and hence what gets maintained—is the product of institutional attention, not of technical fact.

A subsidiary observation: the staff account makes plain that **adoption metrics themselves are a visibility instrument**, not just a passive measurement. "Interaction" usage data is the easiest signal to perform—and teachers perform it under administrative pressure—meaning that what is being measured at a flagship site and what is being measured at the Tier-3 limit case may differ in kind, not just in degree. The dashboard can read **dispositions to perform adoption** at high-attention sites and **dispositions to triage scarce resources** in the thinly-covered limit case

³⁵TeacherT01, Module-C teacher interview, 02:40.

³⁶TeacherT04, Module-C teacher interview, 17:42.

³⁷Paraphrased from S01's staff interview.

³⁸Field notes from primary corpus period; categorization terms surfaced during fieldwork conversations with on-site staff.

³⁹T10/staff exchange paraphrased; field notes from primary corpus period.

⁴⁰TeacherT10, Module-C teacher interview, 37:11.

as equivalent units of “adoption.” That equivalence is part of how the visibility regime sustains itself.

5 Discussion

5.1 Re-framing the “Teacher Adoption” Vocabulary

The “teacher adoption” vocabulary, in its current operationalization in industry and in much of the EdTech research literature, treats the deployed system as a stable object whose uptake is a property of the teacher. The findings of this paper show that this vocabulary fails to see three things—each at a different analytical level.

First, it fails to see the deployed system as **structurally incomplete**. The corpus documents three registers in which the system assumes conditions—time-budget elasticity, substrate reliability, and content substitutability—that classroom practice does not provide (§4.1, §4.2, §4.3); each requires continuous physical and material repair labor to bridge. The frame reads this bridging labor as “use of system features,” attributing to teacher behavior what is in fact compensation for structural gaps.

Second, it fails to see that adoption itself is **multi-actor mediated**. What dashboards record as the teacher’s interaction event is the visible end-point of cognitive and relational labor distributed across roles—subject-group heads articulating design proposals, vice principals manufacturing institutional time, on-site staff translating between vendor and teacher, administrators routing access (§4.4). This mediation differs in kind from material repair: it organizes the relations and timing through which the system reaches the classroom at all. The frame reads multi-actor mediation as “individual teacher adoption.”

Third, it fails to see that visibility itself is **unevenly distributed**. The same breakdown categories appear at flagship and thinly-covered sites, but the routing conditions under which they become product-visible differ systematically (§4.5). This is not a difference in labor distribution but a difference in **visibility regime**: the deployed measurement infrastructure renders labor legible or absorbed depending on commercial attention allocation. The frame reads tier-asymmetric visibility as “differences in adoption rates across schools.”

These failures share one inversion: tensions between deployed system, distributed repair labor, and measurement infrastructure are rewritten by the adoption frame as properties of the teacher. System-design gaps become teacher motivation; collective coordination becomes individual teacher behavior; routing-driven visibility becomes engagement difference. My proposal is not to abandon adoption but to **de-center** it: to treat adoption metrics as one visible surface of a larger installed base [50] composed of material repair labor, multi-actor mediation, and the visibility regime that distributes both.

This positions the paper alongside a recent line of HCI/CSCW scholarship on the hidden labor of AI integration [18, 33], on ghost work and the human-labor infrastructure that AI systems require to function [21, 55], and on the political economy of platform pedagogies (§2.4). What this paper adds is the routing structure through which structural incompleteness becomes someone’s daily work: a vendor-attention hierarchy, a multi-actor mediation chain, and a four-form labor taxonomy specific to deployed K–12 EdTech.

5.2 Infrastructure Repair as an HCI Analytic for Deployed K–12 EdTech

Harvey et al. [24] identify a gap between industry framings of AI-in-education and HCI accounts of teacher–AI collaboration. This paper’s contribution advances that conversation by naming a specific mechanism—repair—that operates at the joints of the gap, and by adding the **on-site staff** layer to the analysis. Three research moves an infrastructure-repair lens enables:

Moving from “teacher adoption” to “human–platform infrastructural coupling.” The analytical question shifts from whether teachers adopt the system to how the system’s, the teachers’, and staff members’ work-tempos, content-pipelines, and coordination structures are coupled, and where the coupling breaks. This is a relational analysis the adoption literature is not equipped to conduct. The coupling is moreover multi-actor by structure, not dyadic between teacher and system: **mediation analysis**—following the chain of subject-group heads, vice principals, on-site staff, and administrators that produces dashboard-visible engagement—becomes the appropriate methodological orientation.

Surfacing invisible repair labor through dashboard redesign. A repair-informed dashboard would ask not “what do teachers do on the platform” but “what do teachers, students, and on-site staff do to make the platform work.” This might include tracking workarounds, content-edit-before-publish events, customer-support tickets by school type, on-site-staff time allocation, and the ratio of platform-recommended to teacher-substituted content. Underlying these proposals is a recognition that the dashboard is not a neutral measurement instrument: by selecting what to count and whose work to count, it constructs the very category “adoption” that the design literature seeks to evaluate. Dashboard redesign is, in this sense, a recategorization of adoption itself. This direction aligns with human-centred and participatory learning-analytics work that treats dashboard design as a situated negotiation with teachers rather than a one-way act of data visualization [45, 59]. The methodological challenge is that much of this labor is by definition off-platform; the design challenge is to construct visibility of it without importing surveillance dynamics that would themselves break the human–system relation.

Interrogating the political economy of visibility. The **visibility regime** documented in §4.5 is unevenly distributed across vendor-attention tiers and has distributive consequences for human labor. This connects to a broader line in critical EdTech [47, 60] about how data infrastructures encode and reproduce institutional hierarchies, and to recent calls in CSCW to *study up* the data infrastructures and labor regimes behind machine-learning systems rather than only studying their downstream effects on users [35]. An HCI research agenda on deployed K–12 EdTech would benefit from taking this seriously: what does it mean to design a product whose feedback loops under-represent certain deployment sites even while counting them as adopted?

5.3 The Rhetorical Work of “AI” / “Smart” Labels: Marketing Substance, Operational Form

§4.3 found that the deployed system’s outputs deliver **form**, while teachers supply the **substance** that contextual judgment requires. The same form/substance divide governs how the deployed suite is *marketed*. Brand labels—“AI-augmented teaching,” “smart classroom,” “intelligent grading”—promise substance; operational delivery, as documented above, is form. Although these labels appear to designate distinct technical categories—machine-learning grading vs digital simulation vs interactive whiteboard—in this Vendor’s product framings they are bundled together under “AI-augmented” / “smart classroom” as a unified marketing brand. I read this gap not as accidental imprecision but as **rhetorical work**: the labels do something, and what they do has consequences for how repair labor is rendered analytically and politically invisible [9, 10, 37, 42]. The rhetorical work has three effects on visibility.

The first effect is **capability inflation**: the marketing label invites attribution of substantive capability to operational form. The issue is not whether AI exists in the stack (it does, in mixed form; §3.1) but whether the operational channels through which components deliver substance match what the label invites users to expect. The corpus shows four registers.

Operational narrowness underneath panel breadth. T09 reported that across a full semester of teaching Module-A, only one of four “AI training models” on the platform’s AI-Innovation-Education panel was usable in classroom deployment.⁴¹

Demoed-but-withdrawn substance. TeacherT01 saw a third-party physics simulation lab in a Module-C demo lesson and formed an expectation around it; when she searched for it in her own deployment, the feature had been withdrawn.⁴²

Surface use inside substance-promising features. TeacherT03 characterized her Module-C use as “surface-level interaction use”—features deliver “emotional value” but do not contribute to “deeper subject learning.”⁴³

Form-mimicking without substance. The Module-H per-student error booklet is marketed as AI-driven personalization but operationalizes a long-standing teacher-driven practice—the student error notebook—only at the form layer; the substance (selection logic against student misconception) does not survive the transposition. T08’s “careless packaging” diagnosis (§4.3) is the user-side articulation.

Capability inflation does not require AI absence: T02’s speech-recognition feature delivered substantial labor savings in her English class.⁴⁴ The inflation is in the breadth of the marketing label relative to the narrowness of the channels in which substance is operationally delivered—what Raji et al.’s [42] critique of AI functionality calls asking “whether it works as claimed” before asking whether it is beneficial.

The second effect is **labor displacement**: when a system is read as “AI-powered,” the substance teachers continue to supply is read as system performance rather than as teacher labor. This resonates with HCI/CSCW work on data cascades and on the broader human-labor infrastructure that AI systems require to function [8, 44], which shows how undervalued data and maintenance work surface later as downstream system failure. TeacherT06’s afternoon labor of swapping in legacy question-bank items, TeacherT07’s hand-grading of free-response answers, the staff member’s daily tablet cabling—these are read by the visibility regime as instances of adoption. T09 puts the displacement in plainer terms: when Module-A’s curriculum content is thin, with weaker-prepared students “everything depends on the teacher.”⁴⁵ Naming the operational stack more precisely would shift the readability of this labor from “use of an AI system” to “manual provision of the substance the algorithmic form does not generate.”

The third effect is **accountability deflection**: the label’s substance-promise can displace responsibility for system behavior away from designers, procurement decisions, and institutional deployment choices toward the abstract category of “the AI”—or, in a mirroring direction, toward the teacher herself. Elish’s [16] account of moral crumple zones describes the first direction: proximate human operators absorb responsibility for automated-system failures despite having limited control over system design. T09’s self-attribution of Module-A’s content thinness to her own under-preparation (“I haven’t put the time in to dig deeper”)⁴⁶ displays the second direction. Both move design responsibility out of the design conversation. Teachers in this corpus often identified the actual mismatches clearly—tagging schema, question-bank coverage, difficulty parameters, classroom time budget—but the institutional label makes those identifications harder for product teams and administrators to treat as design responsibilities. This dynamic operates within the

⁴¹T09, AI Innovation Education teacher interview, approximately 24:03–24:33. The four panel models were text classification, face recognition with style effects, automated classical-poetry generation, and voice synthesis.

⁴²TeacherT01, Module-C teacher interview, 12:03–16:10.

⁴³TeacherT03, classroom teacher interview, 03:00–04:23.

⁴⁴TeacherT02, Module-C teacher interview, 07:30 / 23:11.

⁴⁵T09, AI Innovation Education teacher interview, 13:49–16:33.

⁴⁶T09, AI Innovation Education teacher interview, approximately 16:00.

broader political economy of algorithmic systems Burrell and Fourcade [7] describe as the consolidation of authority over what gets built into a ‘coding elite’ that extracts value from a marginalized labor base, and within the corporate capture of AI knowledge production Whittaker [58] flags as a structural condition shaping HCI’s own analytical horizons.

These three effects together connect to design-ethics scholarship on UX patterns and rhetorical framings [2, 20]. The analytical tradition that attends to “dark patterns” and to the rhetorical construction of user-interface elements is well-positioned to extend its reach upward—to the framings under which entire deployed systems are marketed and procured. The “AI” / “smart” label, as deployed in the Chinese K–12 EdTech market observed, functions as a **system-level rhetorical pattern**: it organizes, at scale, which kinds of labor become legible as system performance and which become invisible. In this sense the labels are part of the visibility regime developed in §4.5—they do measurement work prior to any dashboard.

The argument does not depend on a sharp definitional line for “AI” or “smart”; it depends on the gap between operationally-delivered form and rhetorically-promised substance. That gap is observable independent of definitional precision.

5.4 Limitations

The study has limitations that must be held explicitly in view, organized below by methodology, corpus scope, and voice coverage.

Methodological. The corpus is built from retrospective interpretive reading of industry field artifacts rather than from formal academic ethnography designed under prospective human-subjects review. The author’s prior Vendor employment shapes the analytical frame; §3.2 and §3.5 document the constraints this places on interpretation, the mitigations applied (auditable source markers, refused framings, paraphrase of high-risk passages), and the residual ethical surface that retrospective analysis cannot eliminate. The technical disclosure in §3.1 is based on triangulated observation rather than full architectural audit.

Corpus scope. The corpus is from a single Vendor and a single suite; cross-Vendor comparative work would strengthen the analysis considerably. The staff analysis relies primarily on one on-site delivery staff voice whose work crossed Tier-1 and Tier-3 schools; I use S01 to analyze a role and routing mechanism visible in this deployment, not to generalize across all vendor staff. The Tier-3 evidence is thin by design and by access; School-S8 / T10 functions as a limit case rather than a representative sample, and the main cross-tier comparison rests on the stronger Tier-1 / Tier-2 evidence.

Voice coverage. The four cities covered are predominantly central and eastern Chinese urban and peri-urban contexts; rural-school deployments, western-region deployments, and non-Han minority contexts are absent—and these are precisely the settings where visibility asymmetry would be expected to operate most strongly. The corpus is also adult-centered: student and parent voices are visible only indirectly through teacher, administrator, and staff accounts. A full account of what deployed K–12 EdTech does requires direct engagement with these perspectives.

6 Conclusion

I have argued that teachers and on-site staff in one Chinese K–12 EdTech deployment are better understood not as adopters of AI technology but as **repair workers** whose sustained, invisible labor makes these systems function. Drawing on a multi-source qualitative corpus, I identified four forms of this repair labor (temporal stitching, defensive redundancy, content re-verification, mediation labor), developed *visibility asymmetry* as the structural condition under which similar breakdowns become product-visible at unequal rates across vendor-attention tiers, and read the “AI” / “smart” labels as part of the same visibility regime that renders this labor invisible. The “teacher

adoption” frame encodes defaults—stable system, friction in the teacher, dashboard-visible activity as the relevant object—that compound this invisibility; re-framing deployed K–12 EdTech through infrastructure, repair, and AI-label rhetoric opens space for an HCI agenda that takes seriously the relational structure these systems require.

Future work should extend this analysis across vendors, across deployment contexts (including rural and minority-region schools), and across the other humans involved: students, parents, staff in other regions, and product-side workers whose own work is shaped by the dashboards questioned here.

Data Availability and AI Use Disclosure

Raw data cannot be publicly shared because the corpus contains non-public workplace field artifacts and potentially identifying information about teachers, schools, vendor staff, and deployment sites. The paper provides anonymized excerpts and aggregate descriptions sufficient to support the analysis; additional materials would require separate ethical and confidentiality review.

Generative AI tools were used for language-level editing, consistency checks, and revision suggestions. The author retained responsibility for all claims, analysis, source selection, and reference verification; no AI-generated text was accepted without author review.

References

- [1] America Education Research Association et al. 2011. Code of ethics: American educational research association. *Educational Researcher* 40, 3 (2011), 145–156.
- [2] Jeffrey Bardzell and Shaowen Bardzell. 2013. What is “critical” about critical design?. In *Proceedings of the SIGCHI conference on human factors in computing systems*. 3297–3306.
- [3] Avital Binah-Pollak. 2025. “There Is Only One Childhood”: A New Interpretation of China’s “Crackdown” on Education Companies. *Modern China* 51, 2 (2025), 205–232.
- [4] Geoffrey C Bowker and Susan Leigh Star. 2000. *Sorting things out: Classification and its consequences*. MIT press.
- [5] Virginia Braun and Victoria Clarke. 2006. Using thematic analysis in psychology. *Qualitative research in psychology* 3, 2 (2006), 77–101.
- [6] Virginia Braun and Victoria Clarke. 2019. Reflecting on reflexive thematic analysis. *Qualitative research in sport, exercise and health* 11, 4 (2019), 589–597.
- [7] Jenna Burrell and Marion Fourcade. 2021. The society of algorithms. *Annual review of sociology* 47, 1 (2021), 213–237.
- [8] Antonio A Casilli. 2025. Waiting for robots: The hired hands of automation. In *Waiting for robots*. University of Chicago Press.
- [9] Kate Crawford. 2021. *Atlas of AI: Power, Politics, and the Planetary Costs of Artificial Intelligence*. Yale University Press, New Haven.
- [10] Kate Crawford and Vladan Joler. 2018. *Anatomy of an AI system: The Amazon Echo as an anatomical map of human labor, data and planetary resources*. AI Now Institute.
- [11] Larry Cuban. 2001. *Oversold and underused: Computers in the classroom*. Harvard university press.
- [12] Arlene Kaplan Daniels. 1987. Invisible work. *Social problems* 34, 5 (1987), 403–415.
- [13] Fred D Davis. 1989. Perceived usefulness, perceived ease of use, and user acceptance of information technology. *MIS quarterly* 13, 3 (1989), 319–340.
- [14] Mathias Decuyper, Emiliano Grimaldi, and Paolo Landri. 2021. Introduction: Critical studies of digital education platforms. 16 pages.
- [15] Jérôme Denis and David Pontille. 2015. Material ordering and the care of things. *Science, Technology, & Human Values* 40, 3 (2015), 338–367.
- [16] Madeleine Clare Elish. 2019. Moral crumple zones: Cautionary tales in human-robot interaction (pre-print). *Engaging Science, Technology, and Society (pre-print)* (2019).
- [17] Diana E Forsythe. 1999. “It’s just a matter of common sense”: Ethnography as invisible work. *Computer Supported Cooperative Work (CSCW)* 8, 1 (1999), 127–145.
- [18] Sarah E Fox, Samantha Shorey, Esther Y Kang, Dominique Montiel Valle, and Estefania Rodriguez. 2023. Patchwork: The hidden, human labor of AI integration within essential work. *Proceedings of the ACM on Human-Computer Interaction* 7, CSCW1 (2023), 1–20.

- [19] Stephen Graham and Nigel Thrift. 2007. Out of order: Understanding repair and maintenance. *Theory, culture & society* 24, 3 (2007), 1–25.
- [20] Colin M Gray, Yubo Kou, Bryan Battles, Joseph Hoggatt, and Austin L Toombs. 2018. The dark (patterns) side of UX design. In *Proceedings of the 2018 CHI conference on human factors in computing systems*. 1–14.
- [21] Mary L Gray and Siddharth Suri. 2019. *Ghost work: How to stop Silicon Valley from building a new global underclass*. Harper Business.
- [22] Kalervo N Gulson, Sam Sellar, and P Taylor Webb. 2022. *Algorithms of education: How datafication and artificial intelligence shape policy*. U of Minnesota Press.
- [23] Ariel Han, Xiaofei Zhou, Zhenyao Cai, Shenshen Han, Richard Ko, Seth Corrigan, and Kylie A Peppler. 2024. Teachers, parents, and students’ perspectives on integrating generative AI into elementary literacy education. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*. 1–17.
- [24] Emma Harvey, Allison Koenecke, and Rene F Kizilcec. 2025. "Don't Forget the Teachers": Towards an Educator-Centered Understanding of Harms from Large Language Models in Education. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. 1–19.
- [25] Thomas Hillman, Annika Bergviken Rensfeldt, and Jonas Ivarsson. 2020. Brave new platforms: a possible platform future for highly decentralised schooling. *Learning, media and technology* 45, 1 (2020), 7–16.
- [26] Kenneth Holstein and Vincent Aleven. 2022. Designing for human–AI complementarity in K-12 education. *Ai Magazine* 43, 2 (2022), 239–248.
- [27] Kenneth Holstein, Bruce M McLaren, and Vincent Aleven. 2019. Designing for complementarity: Teacher and student needs for orchestration support in AI-enhanced classrooms. In *International conference on artificial intelligence in education*. Springer, 157–171.
- [28] Lara Houston, Steven J Jackson, Daniela K Rosner, Syed Ishtiaque Ahmed, Meg Young, and Laewoo Kang. 2016. Values in repair. In *Proceedings of the 2016 CHI conference on human factors in computing systems*. 1403–1414.
- [29] Steven J Jackson. 2014. 11 Rethinking repair. *Media technologies: Essays on communication, materiality, and society* (2014), 221–39.
- [30] Erin Kenneally and David Dittrich. 2012. The menlo report: Ethical principles guiding information and communication technology research. *Available at SSRN 2445102* (2012).
- [31] Jeremy Knox. 2020. Artificial intelligence and education in China. *Learning, Media and Technology* 45, 3 (2020), 298–311.
- [32] Annette Markham and Elizabeth Buchanan. 2012. Ethical decision-making and internet research: Recommendations from the AoIR Ethics Working Committee (Version 2.0). (2012).
- [33] Alexandra Mateescu and Madeleine Elish. 2019. AI in context: the labor of integrating new technologies. (2019).
- [34] Shannon Mattern. 2018. Maintenance and care. *Places Journal* (2018).
- [35] Milagros Miceli, Julian Posada, and Tianling Yang. 2022. Studying up machine learning data: Why talk about bias when we mean power? *Proceedings of the ACM on human-computer interaction* 6, GROUP (2022), 1–14.
- [36] Laura Nader. 1972. Up the anthropologist: Perspectives gained from studying up. (1972).
- [37] Simone Natale and Andrea Ballatore. 2020. Imagining the thinking machine: Technological myths and the rise of artificial intelligence. *Convergence* 26, 1 (2020), 3–18.
- [38] Julian E Orr. 2016. *Talking about machines: An ethnography of a modern job*. Cornell University Press.
- [39] Carlo Perrotta, Kalervo N Gulson, Ben Williamson, and Kevin Witzenberger. 2021. Automation, APIs and the distributed labour of platform pedagogies in Google Classroom. *Critical Studies in Education* 62, 1 (2021), 97–113.
- [40] Volkmar Pipek and Volker Wulf. 2009. Infrastructuring: Toward an integrated perspective on the design and use of information technology. *Journal of the Association for Information Systems* 10, 5 (2009), 1.
- [41] Jean-Christophe Plantin, Carl Lagoze, Paul N Edwards, and Christian Sandvig. 2018. Infrastructure studies meet platform studies in the age of Google and Facebook. *New media & society* 20, 1 (2018), 293–310.
- [42] Inioluwa Deborah Raji, I Elizabeth Kumar, Aaron Horowitz, and Andrew Selbst. 2022. The fallacy of AI functionality. In *Proceedings of the 2022 ACM conference on fairness, accountability, and transparency*. 959–972.
- [43] Daniela K Rosner and Morgan Ames. 2014. Designing for repair? Infrastructures and materialities of breakdown. In *Proceedings of the 17th ACM conference on Computer supported cooperative work & social computing*. 319–331.
- [44] Nithya Sambasivan, Shivani Kapania, Hannah Highfill, Diana Akrong, Praveen Paritosh, and Lora M Aroyo. 2021. "Everyone wants to do the model work, not the data work": Data Cascades in High-Stakes AI. In *proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. 1–15.
- [45] Juan Pablo Sarmiento and Alyssa Friend Wise. 2022. Participatory and co-design of learning analytics: An initial review of the literature. In *LAK22: 12th international learning analytics and knowledge conference*. 535–541.
- [46] Kjeld Schmidt and Liam Bannon. 1992. Taking CSCW seriously: Supporting articulation work. *Computer supported cooperative work (CSCW)* 1, 1 (1992), 7–40.
- [47] Neil Selwyn. 2021. *Education and technology: Key issues and debates*. Bloomsbury Publishing.

- [48] Neil Selwyn, Thomas Hillman, Annika Bergviken-Rensfeldt, and Carlo Perrotta. 2023. Making sense of the digital automation of education. *Postdigital Science and Education* 5, 1 (2023), 1–14.
- [49] Susan Leigh Star. 1999. The ethnography of infrastructure. *American behavioral scientist* 43, 3 (1999), 377–391.
- [50] Susan Leigh Star and Karen Ruhleder. 1994. Steps towards an ecology of infrastructure: complex problems in design and access for large-scale collaborative systems. In *Proceedings of the 1994 ACM conference on Computer supported cooperative work*. 253–264.
- [51] Susan Leigh Star and Anselm Strauss. 1999. Layers of silence, arenas of voice: The ecology of visible and invisible work. *Computer supported cooperative work (CSCW)* 8, 1 (1999), 9–30.
- [52] Anselm Strauss. 1985. Work and the division of labor. *Sociological quarterly* 26, 1 (1985), 1–19.
- [53] Anselm Strauss. 1988. The articulation of project work: An organizational process. *Sociological Quarterly* 29, 2 (1988), 163–178.
- [54] Lucille Alice Suchman. 2007. *Human-machine reconfigurations: Plans and situated actions*. Cambridge university press.
- [55] Paola Tubaro, Antonio A Casilli, and Marion Coville. 2020. The trainer, the verifier, the imitator: Three ways in which human platform workers support artificial intelligence. *Big Data & Society* 7, 1 (2020), 2053951720919776.
- [56] John Van Maanen. 2011. *Tales of the field: On writing ethnography*. University of Chicago Press.
- [57] Viswanath Venkatesh, Michael G Morris, Gordon B Davis, and Fred D Davis. 2003. User acceptance of information technology: Toward a unified view1. *MIS quarterly* 27, 3 (2003), 425–478.
- [58] Meredith Whittaker. 2021. The steep cost of capture. *Interactions* 28, 6 (2021), 50–55.
- [59] Korah Wiley, Yannis Dimitriadis, and Marcia Linn. 2024. A human-centred learning analytics approach for developing contextually scalable K-12 teacher dashboards. *British Journal of Educational Technology* 55, 3 (2024), 845–885.
- [60] Ben Williamson. 2017. Big data in education: The digital future of learning, policy and practice. (2017).
- [61] Ben Williamson. 2021. Education technology seizes a pandemic opening. *Current History* 120, 822 (2021), 15–20.
- [62] Ben Williamson, Rebecca Eynon, and John Potter. 2020. Pandemic politics, pedagogies and practices: digital technologies and distance education during the coronavirus emergency. 107–114 pages.
- [63] Ben Williamson and Janja Komljenovic. 2023. Investing in imagined digital futures: the techno-financial ‘futuring’ of edtech investors in higher education. *Critical Studies in Education* 64, 3 (2023), 234–249.